

Cracking the Evaluation Paradox

Building AI Quality Assessment Systems That Actually Work

Why high BLEU/ROUGE can hide dangerous errors, and how hybrid evaluation fixes it

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Scenario: An enterprise summarization startup shipped a model with strong benchmark metrics, but users report missing key details and factual errors.

Goal: Compute metrics (BLEU/ROUGE + semantic + perplexity), compare against human judgment, identify failure patterns, and design a deployable hybrid evaluation framework.

Environment note: This workspace cannot resolve external pretrained model downloads, so true BERTScore/perplexity (neural) cannot be executed here. The report includes exact BLEU/ROUGE plus clearly-labeled semantic/fluency proxies, and provides a runnable template for computing true BERTScore/perplexity in the standard lab environment.

Dataset Overview

We evaluate 10 source articles, one reference summary per article, and 3 model versions (V1, V2, V3) producing one summary per article.

Domain	# Articles
health	3
technology	3
business	2
science	2

Deliverable 1: Computed Metrics Analysis

Metrics are computed for each model version against the reference summaries. BLEU/ROUGE are exact; semantic similarity and perplexity are reported as labeled proxies in this environment.

model	BLEU	ROUGE-1	ROUGE-2	ROUGE-L	SemanticSim_TFIDF	Perplexity_trigram
V1	0.049324	0.423474	0.130801	0.328725	0.189003	118.582273
V2	0.487779	0.760217	0.570820	0.700411	0.500691	160.786193
V3	0.120203	0.512926	0.189946	0.385998	0.209331	153.149180

Highest-scoring model per metric (and lowest perplexity):

- BLEU: **V2**
- ROUGE-L: **V2**
- ROUGE-1: **V2**
- ROUGE-2: **V2**
- SemanticSim_TFIDF: **V2**
- Perplexity_trigram: **V1**

Comparison analysis (150–200 words)

Across the three model versions, the traditional overlap metrics mostly favor V2: it achieves the best overall BLEU and ROUGE (1/2/L) scores, indicating that its wording and content align closely with the reference summaries. V1 scores low on BLEU/ROUGE because it is vague and omits key details, so it shares fewer n-grams with the references. V3 sits in the middle overall, but it contains the most dangerous failure mode: it can score well on overlap while being factually wrong. A clear example is Article 1, where V3 has the highest ROUGE-L among models even though it inflates revenue to \$52B (the reference and source state \$45.2B). This shows why metric agreement is not enough: BLEU/ROUGE measure surface similarity, not truthfulness. For semantic quality, we report a TF-IDF cosine similarity proxy (and a trigram-LM perplexity proxy for fluency) due to offline constraints; in a full lab environment, BERTScore and neural perplexity should be computed with a pretrained model. The key discrepancy is that 'best by metrics' can still be unsafe without targeted factuality checks and human review triggers.

Deliverable 2: Manual Quality Assessment (>=5 summaries)

We manually evaluate factual accuracy, completeness, coherence, naturalness, and overall quality (1–5). This exposes issues metrics miss.

article_id	model_version	factual_errors	missing_key_info	coherence_issues	naturalness_problems	overall_rating	notes
1	V3	YES (revenue wrong: \$52B vs \$45.2B)	Minor	No	No	2	High overlap with reference but one critical numeric error.
1	V2	NO	Minor (omits investment detail)	No	No	4	Accurate and concise; slightly less complete than reference.
2	V3	YES (mechanism + year)	YES (study size, HbA1c specifics)	No	No	1	Medically misleading; confident tone increases risk.
2	V1	NO (mostly vague)	YES (numbers, phase, results)	Minor	Slightly robotic	3	Safe but incomplete; fails enterprise needs for key facts.
4	V3	YES (65% vs 55%, coal 2030 vs 2032, adds net-zero 2035)	YES (renewables 75%)	No	No	1	Policy inaccuracies; adds details not in source.
4	V2	NO	NO	No	No	5	Matches reference closely, includes key numbers.

Deliverable 3: Metric Failure Pattern Analysis

Automated metrics fail in predictable ways because they reward similarity, not correctness. First, BLEU and ROUGE are n-gram overlap metrics: they score a summary high when it repeats many of the same words as a reference, even if one crucial fact is wrong. In our data, Model V3 illustrates this perfectly. For Article 1 (Tech earnings), V3 achieves the highest ROUGE-L among the three models, yet it replaces the correct revenue (\$45.2B) with an inflated \$52B. Most tokens still match the reference (company name, “Q3,” “23%,” “operating income,” “data centers”), so ROUGE stays high while factual accuracy collapses.

Second, semantic metrics like BERTScore can still miss factual errors because they focus on meaning similarity at the sentence level, not entailment. A sentence such as “available by 2024” can be semantically close to “available in late 2025” even though the timeline is wrong. In Article 2 (Diabetes drug), V3 invents both the mechanism (“targets insulin production” instead of liver glucose production) and a one-year early availability claim. The text remains fluent and on-topic, so a semantic similarity score can remain deceptively strong.

Third, overlap metrics are weak at detecting omissions. V1 summaries are generally safe but vague; they omit key numbers (patient count, percentage improvements, deadlines), so human readers rate them low for completeness even when they are not ‘incorrect.’ Finally, perplexity primarily measures fluency: a confidently written hallucination can have excellent perplexity while being wrong (Article 6: ‘100% effective in human trials’).

These patterns explain the evaluation paradox: metrics are necessary for scale, but insufficient for trust. The fix is hybrid evaluation: combine overlap/semantic scores with targeted factuality checks (numbers, entities, dates), risk-aware routing (health/finance/policy always reviewed), and lightweight human review on triggered cases.

Three concrete examples (high similarity, low trust)

Article 1: Tech Giant Announces Q3 Earnings

Issue: Numeric factual error: revenue inflated (\$52B vs \$45.2B).

ROUGE-L (V3) = 0.535 | ROUGE-L (V2) = 0.429

Reference

XYZ Corp reported record Q3 earnings of \$45.2 billion, up 23% year-over-year, driven by cloud computing sales growth of 31%. The company exceeded analyst expectations with \$12.8 billion in operating income and plans to invest \$8 billion in new data centers.

Model V3

XYZ Corp's Q3 earnings reached \$52 billion, up 23% year-over-year, with cloud computing contributing to the growth. Operating income of \$12.8 billion surpassed forecasts, and the company announced \$8 billion investment in data centers across multiple continents.

Despite the wrong revenue number, ROUGE-L for V3 on Article 1 is the highest among models.

Article 2: New Diabetes Drug Shows Promise

Issue: Mechanism + timeline hallucination: claims insulin production target and availability by 2024.

ROUGE-L (V3) = 0.244 | ROUGE-L (V2) = 0.744

Reference

Stanford researchers announced successful Phase 3 trials of Glucomin, a new type 2 diabetes drug showing 34% improvement over current treatments. The 18-month study of 2,400 patients demonstrated 1.8% HbA1c reduction with minimal side effects, with FDA approval expected in 2025.

Model V3

A new diabetes medication called Glucomin showed promising results in clinical trials, with patients experiencing significant improvements. The drug, which works by targeting insulin production, will be available by 2024 after FDA approval with minimal side effects reported.

The summary sounds fluent but contains medically important inaccuracies.

Article 4: Climate Summit Reaches Agreement

Issue: Policy distortion: wrong emissions target and coal closure deadline.

ROUGE-L (V3) = 0.395 | ROUGE-L (V2) = 0.642

Reference

World leaders at the Geneva Climate Summit agreed to reduce global emissions by 55% by 2035, with developed nations committing to 60% cuts and \$300 billion annual climate finance. The accord includes coal phase-out by 2032 and achieving 75% renewable energy globally.

Model V3

At the Geneva Climate Summit, 167 countries signed an agreement to cut greenhouse gas emissions by 65% by 2035. The pact includes \$300 billion in annual funding, coal plant closures by 2030, and requirements for developed nations to achieve net-zero by 2035.

High lexical overlap can still occur because most keywords are shared (summit, funding, emissions).

Deliverable 4: Hybrid Evaluation Framework (JSON)

A production-ready hybrid framework combines weighted automated metrics with targeted factuality checks and strategic human review gates.

Framework JSON (excerpt)

```
{
  "version": "1.0",
  "objective": "Catch high-overlap but low-trust summaries (factual errors, omissions) using automated metrics",
  "automated_metrics": {
    "weights_percent": {
      "BLEU": 15,
      "ROUGE_L": 35,
      "ROUGE_1": 15,
      "BERTScore_F1": 35
    },
    "note": "Weights total 100%. In offline environments, BERTScore_F1 can be approximated with semantic similarity",
  },
  "human_review_triggers": {
    "domain_risk": {
      "always_review_domains": [
        "health",
        "business",
        "policy",
        "finance"
      ],
      "reason": "These domains have high cost of error (medical/financial/policy misinformation).",
    },
    "thresholds": {
      "auto_review_if": [
        "ROUGE_L < 0.45",
        "BERTScore_F1 < 0.85",
        "numeric_mismatch_count >= 1",
        "named_entity_mismatch_detected == true",
        "contradiction_flag == true"
      ],
      "auto_reject_if": [
        "critical_factual_error_detected == true",
        "hallucination_rate_estimate > 0.05",
        "toxicity_or_policy_violation == true"
      ]
    }
  },
  "manual_quality_checkpoints": [
    "Factual accuracy: verify all numbers, dates, and named entities against the source article.",
    "Completeness: check that 3\u20135 key points from the source are present.",
    "No hallucinations: flag any claim not supported by the source.",
    "Coherence: summary is logically ordered and readable.",
    "Naturalness: tone is professional and appropriate for enterprise clients."
  ],
  "deployment_gates": {
    "auto_approve": {
      "criteria": [
        "ROUGE_L >= 0.55",
        "BERTScore_F1 >= 0.88",
        "numeric_mismatch_count == 0",
        "domain not in always_review_domains"
      ]
    },
    "needs_review": {
      "criteria": [
        "any human_review_trigger == true",
        "domain in always_review_domains"
      ]
    }
  }
}
```

```

    },
    "reject": {
      "criteria": [
        "any auto_reject_condition == true"
      ]
    }
  },
  "rationale_200_300_words": "Our dataset shows that overlap metrics can reward unsafe summaries. V3 produces a
}
...

```

Rationale (200–300 words)

Our dataset shows that overlap metrics can reward unsafe summaries. V3 produces a high-overlap summary for the earnings article but changes the revenue from \$45.2B to \$52B, a critical factual error that ROUGE does not penalize strongly because most tokens still match. To avoid deploying ‘confidently wrong’ summaries, we keep ROUGE-L as the main coverage signal (35%) and reserve 30% of weight for semantic similarity via BERTScore, which helps when wording diverges but meaning is preserved. BLEU and ROUGE-1 provide additional n-gram signals (15% each) without overfitting to exact phrasing. However, no weighted score can guarantee factuality, so we add targeted triggers: any numeric mismatch (currency, percentages, years) routes to human review, because numbers are frequent failure points and are cheap to detect automatically. We also route high-risk domains (health, finance/business, policy) to review by default because error cost is high. The deployment gates are designed to control review load: only summaries that pass both quality thresholds and factuality checks are auto-approved; borderline cases are reviewed; clear hallucinations are rejected. This framework is deployable because it concentrates human effort on high-impact failures rather than reviewing everything.

Appendix: Submission Files Generated

The following files are generated alongside this PDF (ready for upload if the grader requires separate files):

- computed_metrics_analysis.md
- manual_quality_assessment.csv
- metric_failure_analysis.md
- hybrid_evaluation_framework.json

Appendix: Template for True BERTScore + Neural Perplexity

```
# True metric computation (run in standard lab environment)

# pip install transformers bert-score

from bert_score import score as bert_score
from transformers import AutoTokenizer, AutoModelForCausalLM
import torch

# BERTScore
P, R, F1 = bert_score(predictions, references, lang='en', model_type='roberta-large')
bert_f1 = float(F1.mean())

# Neural perplexity using GPT-2
tok = AutoTokenizer.from_pretrained('gpt2')
mdl = AutoModelForCausalLM.from_pretrained('gpt2').eval()
if tok.pad_token is None:
    tok.pad_token = tok.eos_token

def ppl(text):
    enc = tok(text, return_tensors='pt', truncation=True)
    ids = enc['input_ids']
    with torch.no_grad():
        loss = mdl(input_ids=ids, labels=ids).loss
    return float(torch.exp(loss))

avg_ppl = sum(ppl(t) for t in predictions) / len(predictions)
```